Following this session, students will be able to:

1. Interpret the pharmacokinetic properties of the antihypertensive agents in the context of their clinical significance.
2. Relate the mechanisms of action of the alpha blockers, centrally acting sympatholytics, vasodilators, ganglion blocking agents and adrenergic neuron blocking agents to their cardiovascular effects by giving examples of therapeutic applications.
3. Predict the adverse effects, drug interactions, and contraindications when given the mechanism of action and/or pharmacokinetic properties of the individual drug classes.
4. Select the appropriate antihypertensive agent and monitoring parameters for the individual patient based on the drug's pharmacologic effects when presented with a clinical case vignette.

Effects of BLOCKING alpha-1 receptors on vascular smooth muscle.

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MECHANISM	EFFECT
↓ Peripheral vascular resistance by vasodilation of both resistance and capacitance vessels (arterioles and veins respectively)	↓ Blood pressure ↓ Venous return to the heart ↓ Preload
Decreased arterial pressure →	Compensatory Na^+ and water retention by kidney
Blocking contraction of the capacitance vessels enhances venous pooling →	↓ Ability to maintain BP in upright position → **Orthostatic hypotension** (syncope) Especially prominent with first dose or dose increase — **"first dose effect"** (may persist in some patients)

Practice points:

Alpha blockers should be taken with a diuretic to counteract sodium and water retention.
Check standing and recumbent blood pressure to check for persistent orthostatic hypotension
The first dose should be small and taken at bedtime to reduce the risk of falling.

Therapeutic uses of alpha blockers

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Selective alpha-1 blockers Prazosin Terazosin Doxazosin	• Hypertension (mild-moderate) refractory to first-line agents add-on to diuretic, BB, others less protective in reducing CV morbidity/mortality • Urinary retention due to benign prostatic hyperplasia blocking α_1 receptors relaxes smooth muscle of the bladder neck and prostate, which improves urine flow
Nonselective alpha blockers	
Phenoxybenzamine (irreversible blocker)	• Pheochromocytoma to counteract hypertension / sweating due to excessive catecholamine release. • Urinary outlet obstruction
Phentolamine (competitive antagonist)	• Management of extravasation of vasopressors • Reversal of local anesthesia containing vasoconstrictor (dental use) • Diagnosis of pheochromocytoma (*obsolete*)

Alpha blockers adverse effects (not a complete list)

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EFFECT	MECHANISM
Tachycardia by nonselective alpha blockers	α_2AR block → NE release → stimulate cardiac β receptors → reflex tachycardia

- Selective alpha-1 blockers cause little tachycardia. Norepinephrine (NE) released from nerve endings acts on presynaptic α_2ARs, which inhibits its own release.

EFFECT	MECHANISM
Miosis	Blocking ocular α_1ARs blocks the mydriatic effect of sympathetic stimulation → unopposed muscarinic contraction of iris sphincter muscle → miosis
Nasal stuffiness	Blocking αAR-mediated constriction of nasal mucosa vasculature → nasal congestion
Intraoperative floppy iris syndrome	The cause has not been established.
Drug interactions	Other drugs that affect blood pressure

Clonidine / Methyldopa: Pharmacokinetic Properties

	CLONIDINE	METHYLDOPA
A	70-80% bioavailability	~42% bioavailability
D	Low plasma protein binding	Low plasma protein binding
M	Hepatic (inactive metabolites) enterohepatic circulation	**Substrate of COMT** Metabolism in gut and liver
E	Renal (~50% unchanged drug)	Renal (unchanged drug and metabolites)
t½	12-16 hours	1.5 – 2 hours
Onset	0.5 – 1 hour	4-6 hours (oral and I.V.)
Duration	6-10 hours (dosing 2x daily)	24-48 h (dosing 2-3x daily)

Values refer to immediate release oral forms, adults, normal renal function

Clonidine dosage forms: Oral tablets, extended release capsule, transdermal patch, solution for epidural injection

Alpha-2 agonists reduce sympathetic outflow from CNS.

Agonists may:

1. activate postsynaptic α_2 ARs in the brainstem, inhibiting the activity of those neurons,

   and / or

2. activate presynaptic α_2 reducing norepinephrine release,

   and possibly

3. (Clonidine binding to imidazoline receptors may also contribute.)

activation of α_2 adrenergic receptors in the brainstem

↓ sympathetic and

↑ parasympathetic tone

lowers blood pressure
(↓HR and myocardial contractility),

causes bradycardia, and

decreases circulating catecholamine levels

Alpha-2 agonists lower BP by decreasing heart rate, relaxing capacitance vessels and reducing peripheral vascular resistance.

Sensitivity to baroreceptor regulation is maintained.

Renal blood flow and GFR are essentially unchanged.

Clonidine

- Directly activates α_2ARs in brainstem

Clonidine lowers heart rate and cardiac output more than methyldopa does, suggesting the sites of action may not be identical.

Methyldopa

- Taken up into presynaptic vesicles
- Converted to active methylnorepinephrine
- Replaces NE in secretory vesicle
- Stimulates α_2ARs in the brainstem

Alpha-2 Agonists Therapeutic uses

Clonidine	
Refractory hypertension	Add-on after ACEI / ARB, diuretic, and CCB
Pain management	Epidural clonidine prevents pain signal transmission
ADHD	Alternative to SNRI therapy; the mechanism is unknown
Glaucoma	Ophthalmic drops reduce formation of aqueous humor
Substance abuse opioids, alcohol, smoking	Management of catecholamine-induced symptoms
Menopause	Management of vasomotor symptoms (hot flashes, sweating)
Methyldopa	Management of hypertension in pregnancy

Alpha-2 Agonists: Common and potentially serious adverse effects (this is not a complete list)

Patients with cardiovascular disease, i.e., bradycardia, coronary insufficiency, recent MI and conduction disturbances

Severe bradycardia and sinus arrest can occur in patients with SA nodal dysfunction or AV nodal block.

Common	Sedation, drowsiness, mental depression, constipation
Epidural clonidine	Hypotension, respiratory depression

Avoid abrupt discontinuation of alpha-2 agonist. | Gradually taper clonidine.

Sudden discontinuation → catecholamine surge → rebound hypertension headache, apprehension, tremors, abdominal pain, sweating, and tachycardia

Withdraw the beta blocker several days before clonidine is withdrawn.

Removal of clonidine first would allow unopposed vasoconstriction.

Direct vasodilators (oral), hydralazine and minoxidil, for refractory hypertension

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Relaxing the smooth muscle of arterioles

↓

↓ SVR and ↓ MAP

(systemic vascular resistance) ↓ (mean arterial pressure)

compensatory responses

mediated by baroreceptors and the SNS, renin, ANG II, and aldosterone

Vasodilators work best in combination with other antihypertensive drugs that oppose the compensatory cardiovascular responses.	Vasodilator therapy does not cause orthostatic hypotension or sexual dysfunction because sympathetic reflexes are intact.

Vasodilators Therapeutic Uses

Hydralazine	Moderate to severe hypertension Heart failure with reduced ejection fraction: hydralazine + nitrate such as isosorbide dinitrate (evidence of benefit in black individuals) Hypertensive emergency in pregnancy or postpartum 
Minoxidil	Severe hypertension (last line); strong antihypertensive effects
Nitroprusside	Acute hypertension Acute decompensated heart failure Controlled hypotension during surgery
Fenoldopam	Severe hypertension (treatment up to 48 hours) May be used in patients with renal compromise and pediatric patients (short-term)

Vasodilators Class Adverse Effects

- Headache, nausea, anorexia, palpitations, sweating, flushing
- Reflex tachycardia – *may provoke angina or ischemic arrhythmia*
- Salt and water retention – *may lead to development of congestive heart failure*

Beta blocker and diuretic should be co-administered to counteract these compensatory effects.

Note: Because of preferential dilation of arterioles over veins, sympathetic reflexes are intact; postural hypotension and sexual dysfunction are not common problems.

Hydralazine	Lupus-like syndrome – slow acetylators – rash, arthralgia; reversible
Minoxidil	Hypertrichosis – minoxidil topically for hair growth due to male pattern baldness

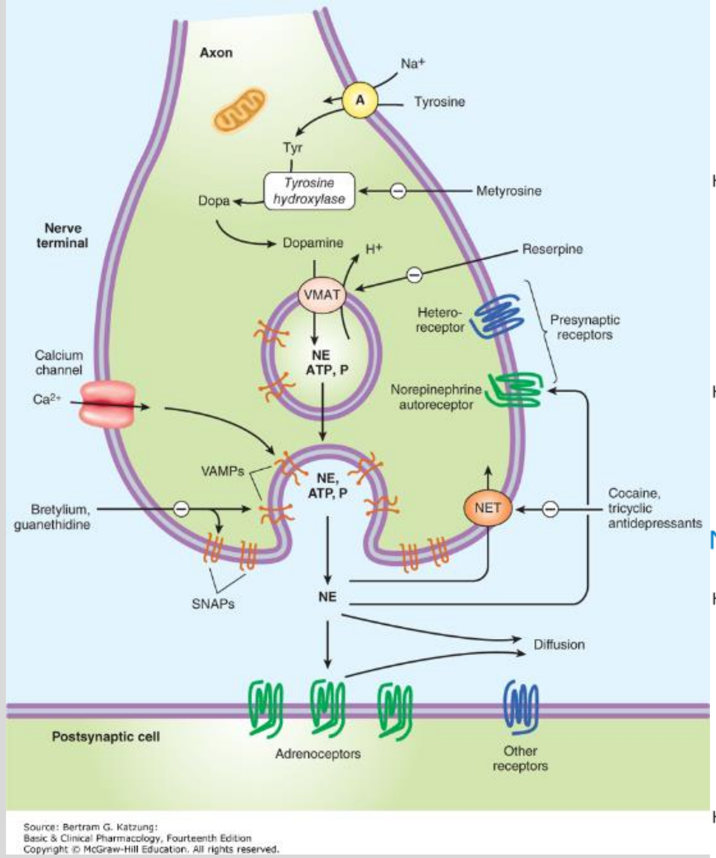
Vasodilators “Unique” Adverse Effects

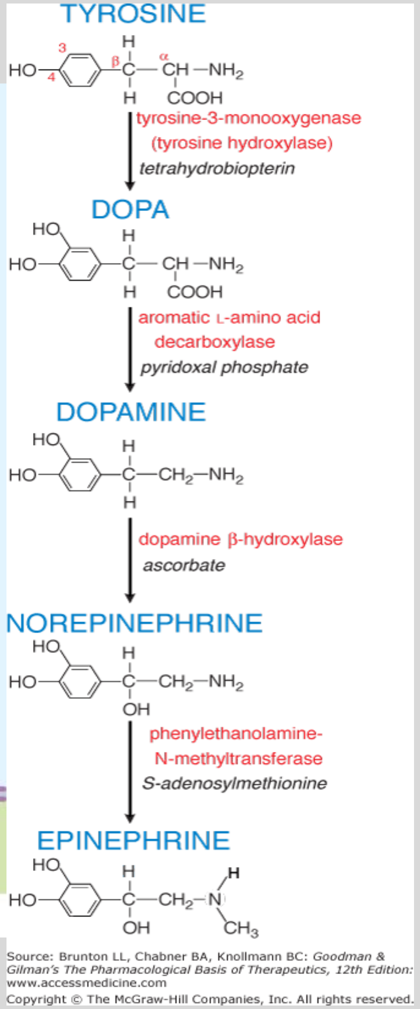
<table>
<tr><td>Hydralazine</td><td colspan="2">Lupus-like syndrome, reversible – rash, arthralgia, myalgia, fever
More likely with prolonged therapy and among slow acetylators</td></tr>
<tr><td>Minoxidil</td><td colspan="2">Hypertrichosis – Topically for hair growth due to male pattern baldness</td></tr>
<tr><td>Nitroprusside</td><td>Excessive BP lowering
Cyanide accumulation</td><td rowspan="2">Antidotes to cyanide accumulation:
1. Sod. nitrite f/b sod. thiosulfate → sulfur donor → cyanide metabolism
2. Hydroxocobalamin → cyanocobalamin (nontoxic vitamin B12)</td></tr>
<tr><td>Fenoldopam</td><td>↑ Intraocular pressure
Avoid in glaucoma patients</td></tr>
</table>

Drugs that cause catecholamine depletion in sympathetic nerve terminals are Reserpine, Guanethidine, Guanadrel, and Metyrosine. They are rarely used and will not be covered here.

Briefly, for your information:

- Reserpine is an irreversible VMAT2 inhibitor, which causes depletion of neurotransmitter centrally and peripherally. Adverse effects include orthostatic hypotension, sexual dysfunction, and sedation and mental depression.
- Guanethidine and guanadrel replace NE in storage vesicles, leading to NE depletion. Neither drug is available in the US.
- Metyrosine blocks tyrosine hydroxylase, the rate limiting step in the biosynthesis of catecholamines, which decreases the levels of endogenous catecholamines.





Left: Katzung & Vanderah, Figure 6-4; Right: Goodman & Gilman, Figure 10-5

Likewise, Ganglion-blocking drugs mecamylamine (oral) and trimethaphan (I.V.) are no longer in use and will not be covered here.

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Briefly for your information:

Ganglion blockers were among the first agents used in the treatment of hypertension. They are no longer available clinically because of intolerable toxicities related to their primary action – blocking ganglionic nicotinic receptors (circled on the figure). The drugs block decrease peripheral resistance and cardiac output by blocking all autonomic outflow.

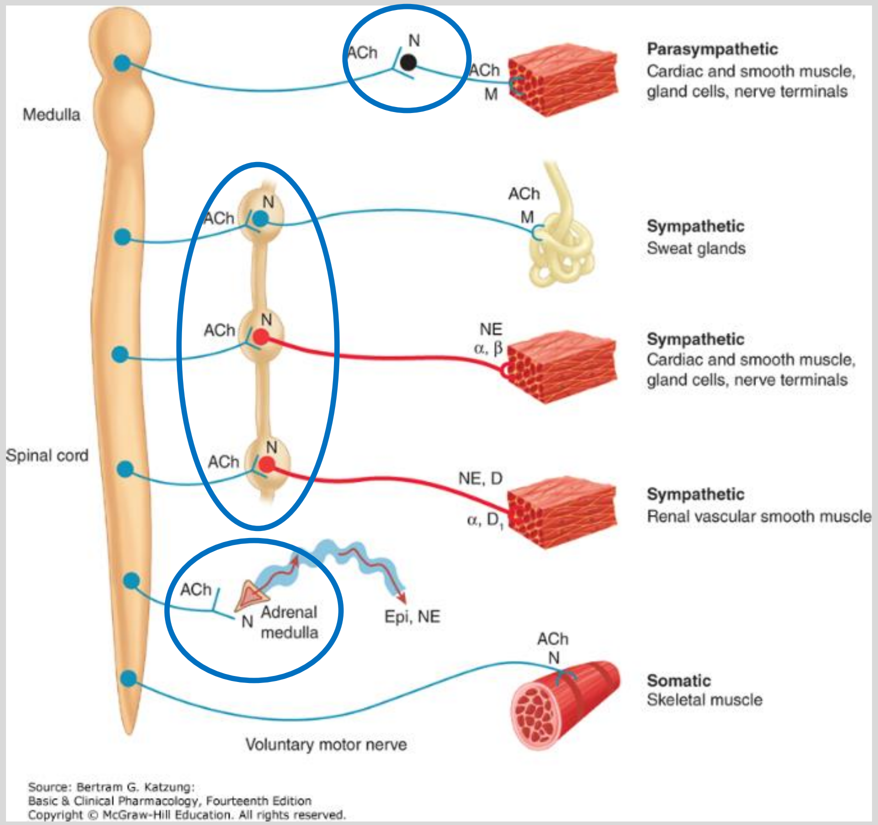
Adverse effects:

– Sympathoplegia: excessive orthostatic hypotension

– Parasympathoplegia: constipation, urinary retention, precipitation of glaucoma, blurred vision / mydriasis, dry mouth

– Sexual dysfunction: and decreased libido, impotence (both erection and ejaculation)

– CNS: altered mental status, choreiform movements, convulsions, fatigue, orthostatic dizziness, paresthesia, sedation



Alpha blockers

- The α_1-adrenergic receptor competitive antagonists (alpha blockers) used in the management of hypertension are prazosin, terazosin, and doxazosin. Terazosin and doxazosin have longer t½ than prazosin and may be taken once daily. They decrease peripheral vascular resistance by causing relaxation of arterial and venous smooth muscle, which reduces preload. The drugs cause only minimal changes to cardiac output, renal blood flow, and glomerular filtration.
- Orthostatic hypotension and reflex tachycardia are common with initiation of therapy – first dose effect. Salt and water retention occur, which is mitigated when the alpha blocker is given with a diuretic.
- Alpha-1 blockers are add-on drugs for the treatment of refractory hypertension. They are not recommended initial treatment due to weak clinical trial data.
- Alpha-1 blockers, terazosin and doxazosin, may also be useful in managing the urinary symptoms of benign prostatic hyperplasia.
- Nonselective alpha blockers are useful in the perioperative management of pheochromocytoma. They cause orthostatic hypotension and tachycardia.

Central sympatholytics: α2-Adrenergic Receptor Agonists

- Alpha-2 agonists inhibit sympathetic vasomotor centers, which reduces sympathetic outflow from the CNS (central sympatholytic effect). Reduction in cardiac output (decreased heart rate) and peripheral vascular resistance and relaxation of capacitance vessels results in lowering of blood pressure. It does not appreciably decrease renal blood flow or glomerular filtration rate.
- Clonidine is administered as oral tablets and long-acting patch for the management of hypertension. Epidural clonidine is used for pain management. Clonidine is also used for managing sympathetic symptoms associated with substance withdrawal and may relieve vasomotor symptoms associated with menopause.
- Abrupt withdrawal can cause rebound hypertension. Beta blockers should be tapered and withdrawn before tapering and withdrawal of clonidine
- Common adverse effects include sedation, dry mouth, bradycardia, and constipation.
- Methyldopa is converted in sympathetic neurons to α-methylnorepinephrine, which displaces norepinephrine in storage vesicles. It activates central α2 receptors. It is safe in pregnancy. Side effects include sedation, bradycardia, and constipation.

Vasodilators

- Direct-acting vasodilators hydralazine and minoxidil relax smooth muscle of the arteries and arterioles, which decreases peripheral vascular resistance, lowering blood pressure. They are used for the management of refractory hypertension. Minoxidil is a powerful antihypertensive agent.
- They cause reflex tachycardia, which could prompt angina, myocardial infarction, and heart failure. They cause salt and water retention. Concurrent beta blocker and diuretic are recommended to counteract these effects.
- Hydralazine with a nitrate is also used in the management of heart failure. Hydralazine is also used in the management of hypertension in pregnancy and postpartum.
- Adverse effects of hydralazine include headache, palpitations, and sweating. Reversible lupus-like syndrome can occur with high dose and poor acetylators (hydralazine is metabolized by NAT2).
- Minoxidil can cause hypertrichosis. Topical minoxidil is used to grow hair.

Vasodilators

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- Nitroprusside is a powerful intravenous vasodilator for the management of hypertensive emergencies. It causes the release of nitric oxide, dilating both venous and arterial blood vessels. Nitroprusside reduces peripheral vascular resistance and venous return. Excessive hypotension can result in death. Nitroprussides blood pressure lowering effects rapidly disappear when the infusion is stopped.
- The most serious toxicity is accumulation of cyanide, which can occur with prolonged infusion or defect in cyanide metabolism. Sodium thiosulfate is a sulfur donor that facilitates cyanide metabolism. Hydroxocobalamin combines with cyanide to form nontoxic cyanocobalamin.
- Fenoldopam is an intravenous dopamine D1 receptor agonist that induces dilation of peripheral arteries and natriuresis. It has a short duration of action (half-life ~10 minutes). It is used for hypertensive emergencies.
- Fenoldopam causes reflex tachycardia, headache, and flushing, and increases intraocular pressure (caution in patients with glaucoma).